

Lineare Algebra
Serie 22
Lukas Mengis

Exercise 1

$$\begin{aligned}\langle x, y \rangle_A &= x^T A y = (x_1, x_2) \begin{pmatrix} a_1 & b \\ b & a_2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = a_1 x_1 y_1 + b x_1 y_2 + b x_2 y_1 + a_2 x_2 y_2 \\ &\Rightarrow a_1 = 1 \quad a_2 = 7\end{aligned}$$

$$A \text{ muss positive definit sein} \Rightarrow a_1 a_2 - b^2 > 0 \Rightarrow |b| < \sqrt{7}$$

Exercise 2

a)

$$\begin{aligned}p, q, s &\in V \quad \lambda \in \mathbb{R} \\ \langle \lambda q + s, p \rangle &= \int_0^\infty \lambda(q + s)(t)p(t)e^{-t} dt = \lambda \int_0^\infty q(t)p(t)e^{-t} dt + \int_0^\infty s(t)p(t)e^{-t} dt \\ \langle p, q \rangle &= \int_0^\infty p(t)q(t)e^{-t} dt = \int_0^\infty q(t)p(t)e^{-t} dt = \langle q, p \rangle \\ \langle p, p \rangle &= 0 = \int_0^\infty p(t)p(t)e^{-t} dt \Rightarrow p = 0\end{aligned}$$

b)

$$\begin{aligned}g_{ij} &= \int_0^\infty t^i t^j e^{-t} dt = (i + j)! \\ \Rightarrow G &= \begin{pmatrix} 0! & 1! & \dots & n! \\ 1! & 2! & \dots & (n+1)! \\ \vdots & \vdots & \ddots & \vdots \\ n! & (n+1)! & \dots & (2n)! \end{pmatrix}\end{aligned}$$

Exercise 3

$$R : V \rightarrow V$$

$$x \mapsto Ax \quad \text{with } A = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

Let $v, w \in V$ then

$$\begin{aligned}\langle Av, Aw \rangle &= (v_1 \cos(\theta) - v_2 \sin(\theta))(w_1 \cos(\theta) - w_2 \sin(\theta)) \\ &\quad + (v_1 \sin(\theta) + v_2 \cos(\theta))(w_1 \sin(\theta) + w_2 \cos(\theta)) \\ &= v_1 w_1 (\cos^2(\theta) + \sin^2(\theta)) + v_2 w_2 (\sin^2(\theta) + \cos^2(\theta)) \\ &= v_1 w_1 + v_2 w_2 \\ &= \langle v, w \rangle\end{aligned}$$

Therefore

$$\begin{aligned}\theta_2 &= \cos^{-1} \left(\frac{\langle Av, Aw \rangle}{\|Av\| \cdot \|Aw\|} \right) \\ &= \cos^{-1} \left(\frac{\langle v, w \rangle}{\|v\| \cdot \|w\|} \right) \\ &= \theta_1\end{aligned}$$

they have the same angle

Exercise 4

(a)

$$(AA^T)^T = A^{TT}A^T = AA^T$$

□

(b)

Eine Matrix ist positiv definit falls gilt

$$\forall x \neq 0 \in \mathbb{R}^n \quad x^T M x > 0$$

angewendet auf $M = AA^T$

$$x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2 > 0$$

Also $Ax \neq 0 \quad \forall x \neq 0 \Leftrightarrow \ker(A) = \{0\} \Leftrightarrow A$ ist invertierbar.

□

(c)

Wir wollen zuerst zeigen das gilt

$$\ker(AA^T) = \ker(A)$$

sei $x \in \ker(A^T A)$

$$\begin{aligned} A^T A x &= 0 \\ \Rightarrow x^T A^T A x &= 0 \end{aligned}$$

da $x^T A^T A x = \|Ax\|^2$ daher

$$x \in \ker(Ax)$$

andere Seite.

sei $x \in \ker(Ax)$

$$x^T A^T A x = x^T A^T \cdot 0 = 0$$

also $\ker(A^T A) = \ker(Ax)$

Es gilt $\dim(\text{im}(A)) = \text{rank}(A)$ daher

$$\text{rank}(A) = \dim(\text{im}(A)) = n - \dim(\ker(A)) = n - \dim(\ker(A^T A)) = \dim(\text{im}(A^T A)) = \text{rank}(A^T A)$$

□

Exercise 5

(a)

$$x, y \in V$$

$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle = \langle x, x \rangle + 2\langle y, x \rangle + \langle y, y \rangle \\ \|x - y\|^2 &= \langle x - y, x - y \rangle = \langle x, x \rangle - 2\langle y, x \rangle + \langle y, y \rangle \\ \|x + x\|^2 + \|x - y\|^2 &= 2\langle x, x \rangle + 2\langle y, y \rangle = 2\|x\|^2 + 2\|y\|^2\end{aligned}$$

□

(b)

1. This norm is always positiv, since it is a sum of absolute values.
2. $v \in V \quad \lambda \in \mathbb{R} \quad \|\lambda v\| = \sum_{i=1}^n |\lambda v_i| = |\lambda| \sum_{i=1}^n |v_i| = |\lambda| \|v\|$
3. $v, w \in V \quad \|v + w\| = \sum_{i=1}^n |v_i + w_i| \leq \sum_{i=1}^n |v_i| + |w_i| = \sum_{i=1}^n |v_i| + \sum_{i=1}^n |w_i| = \|v\| + \|w\|$

Counterexample: using the *parallelogram identity*

$$x_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad x_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\|x_1\| = \|x_2\| = 1$$

$$\|x_1 + x_2\| = \|x_1 - x_2\| = 2$$

$$2^2 + 2^2 = 8 \neq 2 * 1 + 2 * 1 = 4$$

Exercise 6

Let $k = \mathbb{R}, V = m_{n \times n}(K)$

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow K$$

$$(A, B) \mapsto \text{Tr}(A^T B) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ij}$$

Symmetric:

$$\langle A, B \rangle = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ij} = \sum_{i=1}^n \sum_{j=1}^n b_{ij} a_{ij} = \langle B, A \rangle$$

linearity $\lambda \in R \quad A, B, C \in V$

$$\langle \lambda A + C, B \rangle = \sum_{i=1}^n \sum_{j=1}^n (\lambda a_{ij} + c_{ij}) b_{ij} = \lambda \langle A, B \rangle + \langle C, B \rangle$$

Positive definite $A \in V$

$$\langle A, A \rangle = \sum_{i=1}^n \sum_{j=1}^n a_{ij}^2 \quad \forall a_{ij}^2 \geq 0 \quad \Rightarrow \langle A, A \rangle = 0 \Leftrightarrow A = 0$$